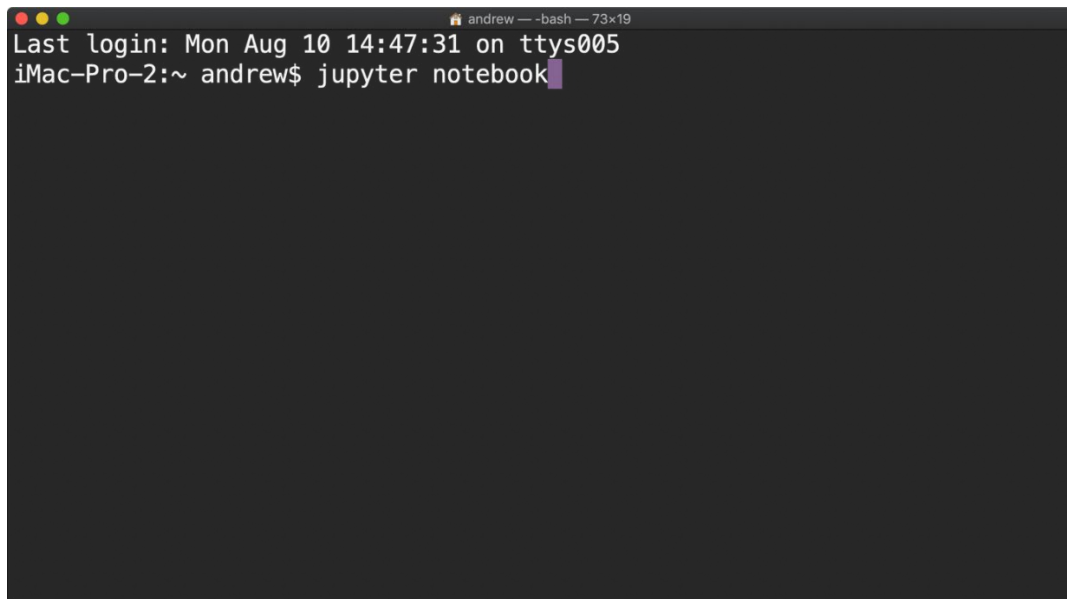


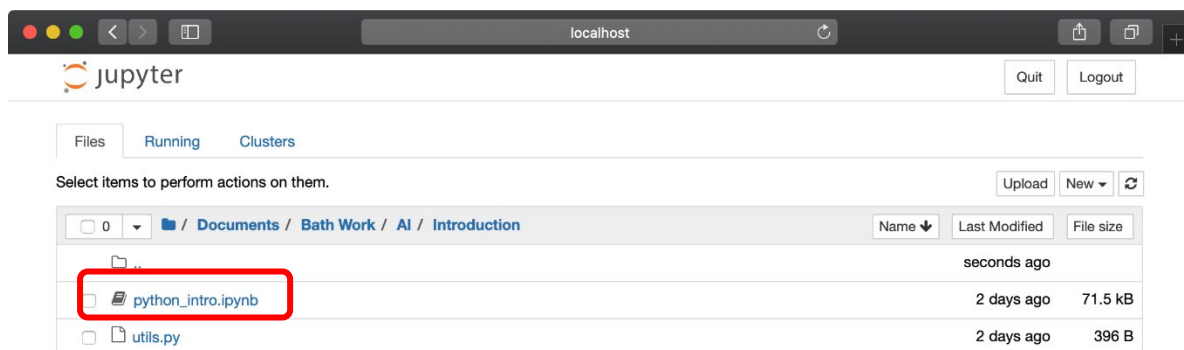
Opening Jupyter files (macOS)

1. Make sure you have:
 - a. installed Anaconda
 - b. downloaded the intro-to-python-jupyter files from Engage
 - c. noted where these files are located on your computer
2. Open the Terminal app. You will find this in Applications, inside the Utilities folder. Alternatively, press Cmd+Space and type "Terminal". Now type `jupyter notebook`, as shown in the screenshot below, then hit enter.

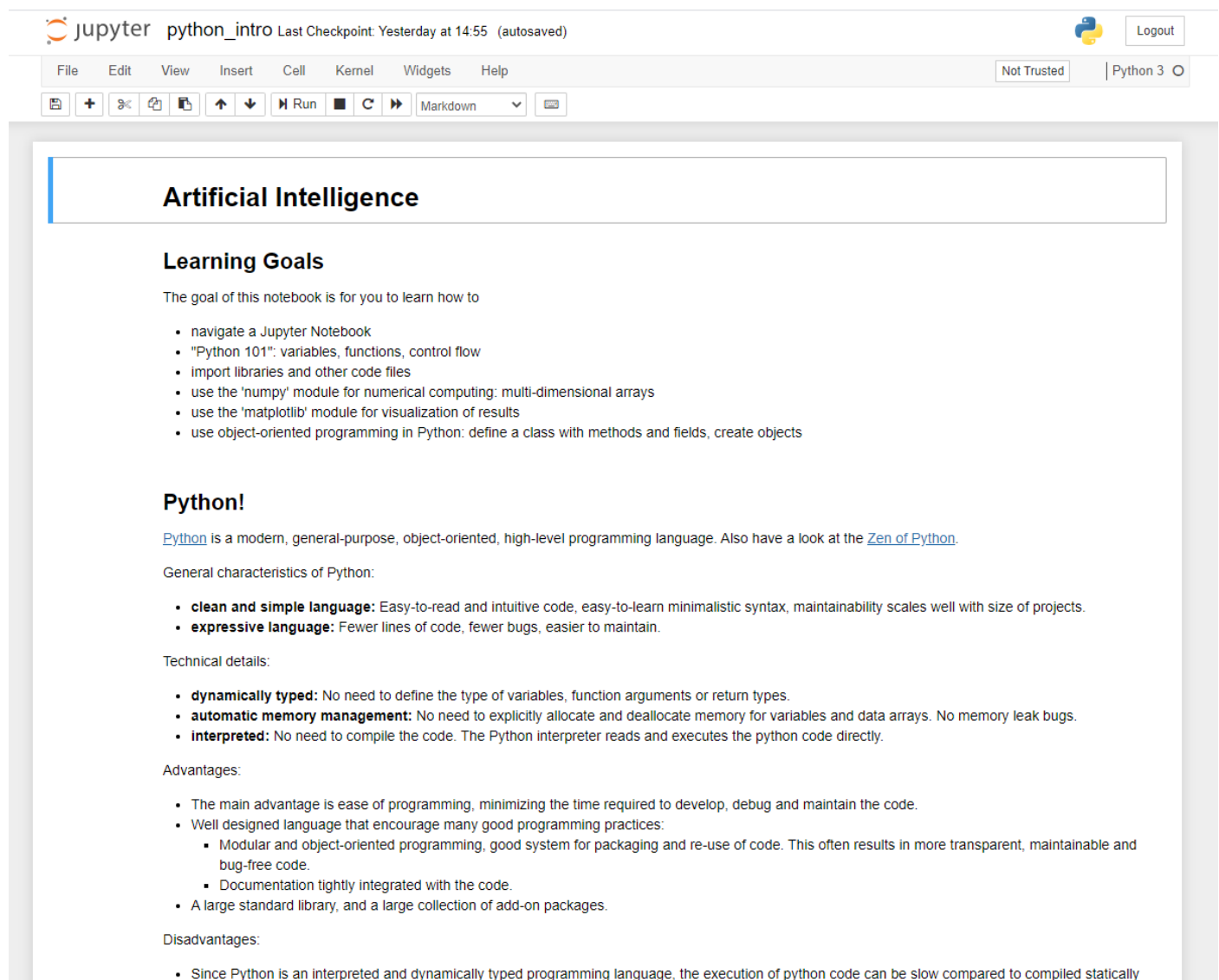


```
andrew — -bash — 73x19
Last login: Mon Aug 10 14:47:31 on ttys005
iMac-Pro-2:~ andrew$ jupyter notebook
```

3. This will launch the Jupyter notebook file interface inside your web browser. Use the folder structure to navigate to the place where you saved the files before, and open the folder called "python_intro.ipynb".



4. Now click on the file "python_intro.ipynb" to open the notebook file.



Jupyter python_intro Last Checkpoint: Yesterday at 14:55 (autosaved) Logout

File Edit View Insert Cell Kernel Widgets Help Not Trusted Python 3

Artificial Intelligence

Learning Goals

The goal of this notebook is for you to learn how to

- navigate a Jupyter Notebook
- "Python 101": variables, functions, control flow
- import libraries and other code files
- use the 'numpy' module for numerical computing: multi-dimensional arrays
- use the 'matplotlib' module for visualization of results
- use object-oriented programming in Python: define a class with methods and fields, create objects

Python!

[Python](#) is a modern, general-purpose, object-oriented, high-level programming language. Also have a look at the [Zen of Python](#).

General characteristics of Python:

- **clean and simple language:** Easy-to-read and intuitive code, easy-to-learn minimalistic syntax, maintainability scales well with size of projects.
- **expressive language:** Fewer lines of code, fewer bugs, easier to maintain.

Technical details:

- **dynamically typed:** No need to define the type of variables, function arguments or return types.
- **automatic memory management:** No need to explicitly allocate and deallocate memory for variables and data arrays. No memory leak bugs.
- **interpreted:** No need to compile the code. The Python interpreter reads and executes the python code directly.

Advantages:

- The main advantage is ease of programming, minimizing the time required to develop, debug and maintain the code.
- Well designed language that encourage many good programming practices:
 - Modular and object-oriented programming, good system for packaging and re-use of code. This often results in more transparent, maintainable and bug-free code.
 - Documentation tightly integrated with the code.
- A large standard library, and a large collection of add-on packages.

Disadvantages:

- Since Python is an interpreted and dynamically typed programming language, the execution of python code can be slow compared to compiled statically